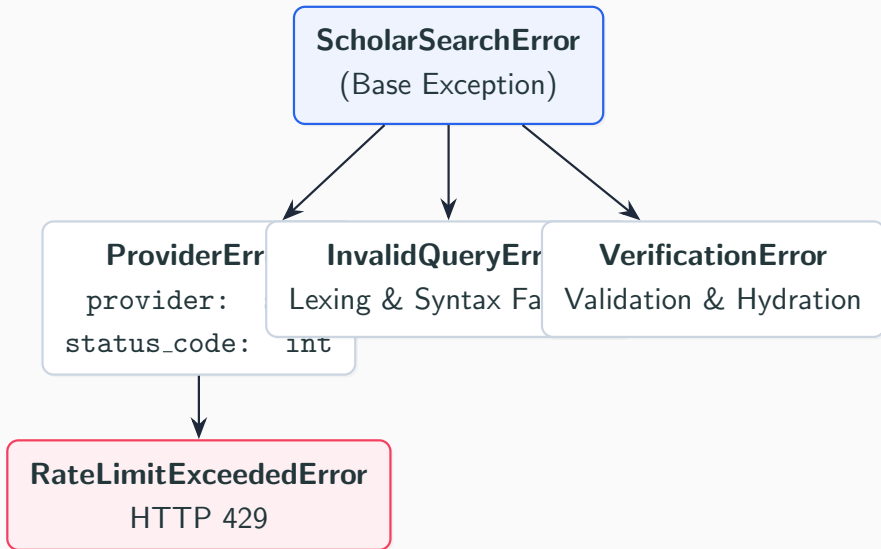


Building a Reproducible Scholarly Search Toolkit

Episode 7: Typed Exception Hierarchy

Nexus Scholar Suite

The Exception Hierarchy



Exception Implementation

```
class ScholarSearchError(Exception):
    """Base exception for all scholar-search errors."""
    pass

class ProviderError(ScholarSearchError):
    """Raised when an external academic provider returns an error."""
    def __init__(self, provider: str, message: str, status_code: int | None = None):
        self.provider = provider
        self.status_code = status_code
        super().__init__(f"[{provider}] {message}" +
                        (f" (HTTP {status_code})" if status_code else ""))

class RateLimitExceededError(ProviderError):
    """Raised when an API rate limit is exceeded (HTTP 429)."""
    def __init__(self, provider: str, message: str = "Rate limit exceeded"):
        super().__init__(provider=provider, message=message, status_code=429)
```